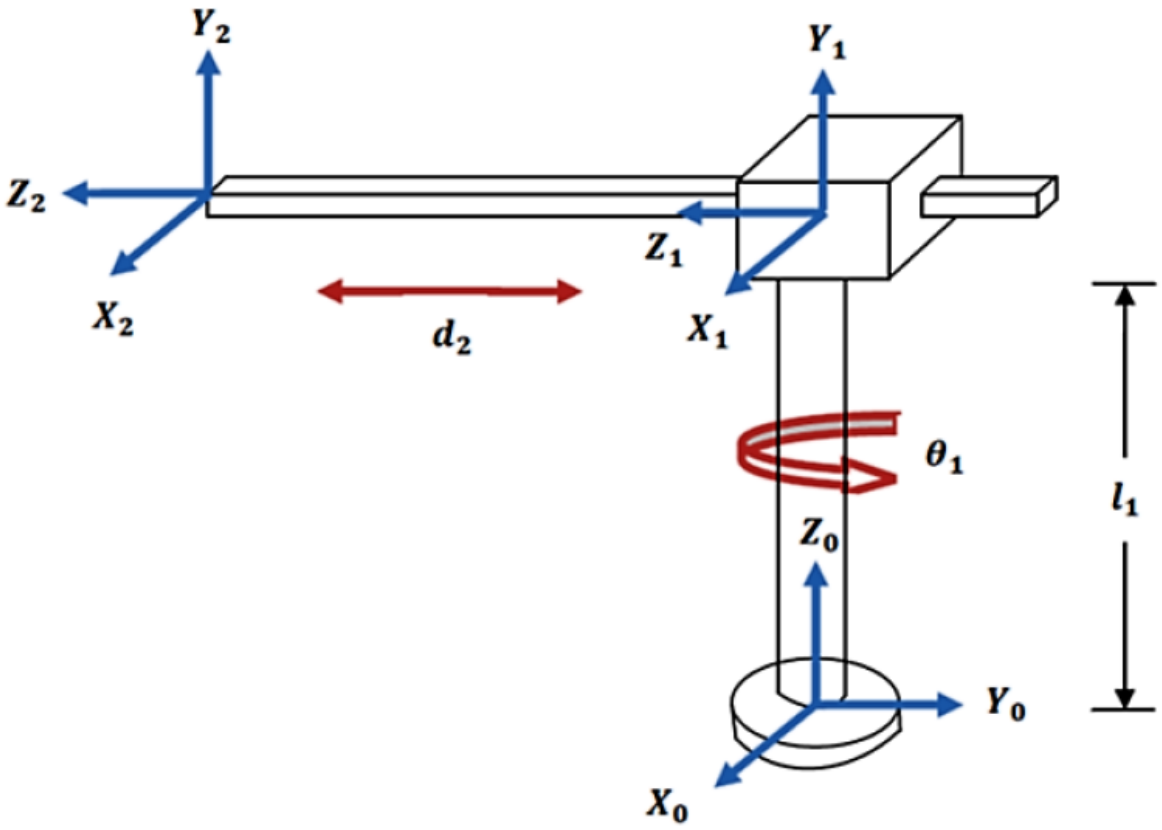


$$
{}^{j-1}A_j = \begin{pmatrix} \cos\theta_j & -\sin\theta_j\cos\alpha_j & \sin\theta_j\sin\alpha_j & a_j\cos\theta_j \\ \sin\theta_j & \cos\theta_j\cos\alpha_j & -\cos\theta_j\sin\alpha_j & a_j\sin\theta_j \\ 0 & \sin\alpha_j & \cos\alpha_j & d_j \\ 0 & 0 & 0 & 1 \end{pmatrix}
$$

$$A_2^0 = A_1^0 * A_2^1$$

$$A_1^0 = \begin{bmatrix} C\theta_1 & 0 & S\theta_1 & 0 \\ S\theta_1 & 0 & -C\theta_1 & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A_2^1 = \begin{bmatrix} 1 & 0 & S\theta_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2^0 = \begin{bmatrix} C\theta_1 & 0 & S\theta_1 & d_2 S\theta_1 \\ S\theta_1 & 0 & -C\theta_1 & -d_2 C\theta_1 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$x = d_2 sin\theta_1$$

$$y = -d_2 cos\theta_1$$

$$z = l_1$$

Inverse Kinematics:

$$\theta_1 = tan^{-1} \frac{-x}{y}$$

$$d_2 = \sqrt{x^2 + y^2}$$